Distinct Dynamics of Parvalbumin-Positive Interneurons in Hippocampal CA1 at the Onset of Extinction

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Abstract

Extinction is the suppression of a learned response when its previously associated outcome no longer occurs. Both original associative learning and subsequent extinction learning depend on the hippocampus formation in the medial temporal lobe. My senior thesis addressed how local inhibitory and excitatory populations in the hippocampus are altered through the process of new learning and subsequent extinction learning in the same animals. To address this question, we performed wide-field single-photon calcium imaging and recorded the local field potential (LFP) in parallel to assess population dynamics in dorsal CA1 of PV-tdTomato-Cre mice during tone–air-puff trace conditioning followed by extinction. We focused on the most abundant inhibitory cell class in the hippocampus: parvalbumin-positive (PV+) interneurons. Both associative learning and extinction were recorded in a single session, so the same cells could be followed across the withdrawal of the tone that served as the predictive unconditioned stimulus (US). Conditioned responding (eyeblink) to a noxious air puff decreased in every animal once the US was withdrawn and this occurred over 10-40 trials. In the four animals with the largest PV+ samples, the identities of highly active PV+ cells showed enhanced shift in responsivity at the transition to extinction training than those of non-PV populations. This effect was most prominent in the first ten extinction trials. In addition, we analyzed the LFP of these four animals from the hippocampus and found that the theta power (9–12 Hz) at tone onset was larger on extinction trials than on conditioning trials. This increase did not appear as a step at the first omitted US but built up over the extinction trials as new learning occurred. These results suggest that the PV+ cell activity dynamics may be a precursor to the network changes associated with extinction learning in the hippocampus.

1 Introduction

1.1 Aversive learning and extinction

Learning adapts the brain to experience by changing cellular connections and networks. In associative conditioning, the brain links a conditioned stimulus (CS) to an unconditioned stimulus (US). Extinction is the gradual reduction of the conditioned response when the CS is repeatedly presented without the US. Extinction learning is thought to reflect a new inhibitory memory that competes with the original association, which is not erased. The CS therefore has two competing meanings, and context influences which one is expressed.[1]

The hippocampus provides this contextual information. It encodes context relevant to the expression of fear or extinction as part of a circuit that also includes the medial prefrontal cortex and amygdala.[2] Such context-specific expression of extinction can be eliminated by reversible inactivation of the dorsal hippocampus.[3] During extinction training, inactivation stops the new memory from being linked to its context thereby delaying extinction demonstrating a role for the hippocampus in both retrieval and extinction learning.[4]

These mechanisms are clinically relevant because exposure-based therapies for anxiety disorders rely on extinction training through exposure based therapies.[2] Patients with post-traumatic stress disorder (PTSD) can acquire and extinguish conditioned fear normally but fail to recall the extinction memory a day later.[5] Dysfunction in the hippocampal contextual circuit has been implicated in PTSD and related disorders.[2] Understanding how the hippocampus shifts from expressing the learned association to expressing extinction may inform these treatments.

1.2 Trace eyeblink conditioning

Trace eyeblink conditioning (tEBC) provides a temporally precise paradigm for studying this learning. A tone CS is separated from an air-puff US by a brief stimulus-free trace interval without a sensory stimulus bridges. Delay conditioning, in which the CS and US overlap, can be supported by brainstem–cerebellar circuitry alone.[6] Trace conditioning with a sufficiently long interval also requires the hippocampus.[7]

CA1 is required for trace conditioning. The task transiently increases the intrinsic excitability of CA1 pyramidal neurons in a learning-specific manner.[8] Optogenetic silencing of dorsal CA1 during the stimulus-free gap disrupts trace conditioning,[9] indicating that CA1 activity across the gap is needed to associate the CS and US to one another. tEBC also provides a defined analysis window, from tone onset to air-puff onset. CA1 activity must maintain information linking the CS and US during this interval, so extinction-related changes in that link may be detectable within it.

1.3 Conditioning and extinction populations in CA1

CA1 represents conditioning and extinction at the single neuron level can be revealed across the population. Mount et al. reported that trace conditioning and extinction recruit two distinct, minimally overlapping neuronal populations[10]: conditioning (CO) cells that respond during conditioning and extinction (EX) cells that emerge during extinction. Similarly, CA1 neurons active during the stimulus-free period of trace fear conditioning are reactivated at retrieval and lose this activity during extinction.[11]

These findings shift the question from whether CA1 represents extinction to how its output shifts from the CO population to the EX population. Local inhibition is one possible mechanism: inhibitory interneurons can rapidly (across a few trials) alter which principal cells fire without

requiring slower synaptic rewiring that would be associated with protein synthesis dependent remodification.

Among CA1 inhibitory interneurons, parvalbumin-expressing (PV+) basket cells could provide such a gate. They fire rapidly and target the perisomatic region of pyramidal cells, allowing them to regulate the timing, gain and synchrony of CA1 output.[12] PV+ networks also show learning-related plasticity. Experience can shift PV basket-cell networks between states that favor or constrain further plasticity, and fear conditioning drives them into a high-PV state.[13]

Two lines of evidence identify PV+ cells as candidates for regulating the conditioning-to-extinction transition in tEBC. First, dorsal hippocampal PV+ interneurons show sustained activity across the CS and trace interval early in tEBC acquisition. This activity is greater on trials with a conditioned response, and optogenetic suppression impairs trace but not delay conditioning.[14] Once conditioned responding reaches asymptote, the sustained activity falls to the level of unpaired controls, and suppressing it no longer affects performance. Thus, PV+ involvement in tEBC may have a time limited role across learning. Second, PV+ interneurons contribute to gamma oscillations. Inhibitory networks pace pyramidal-cell firing into gamma cycles through perisomatic inhibition.[15] Activating PV+ cells can generate gamma rhythms, whereas suppressing them reduces gamma.[16] In CA1, slow and fast gamma couple the region to different inputs, suggesting that gamma frequency may route information between pathways.[17] PV+ interneurons have also been shown to mechanistically contribute to theta (4-12 Hz) rhythms. In an isolated hippocampus, activating PV+ populations at 8 Hz led to robust pyramidal entrainment that was phase synchronized and could be eliminated simply by silencing PV cells.[18] Changes in PV+ activity could therefore affect ensemble activity, communication between CA1 and its inputs, and the population activity that underlines CA1 rhythms.

Recent work links PV+ interneurons and gamma to extinction in other regions. In ventral CA1, lateral entorhinal input to PV+ interneurons drives low-gamma synchronization during fear extinction, and manipulating these cells can alter extinction.[19] In medial prefrontal cortex, PV+ interneurons regulate fear extinction and mediate stress-related impairment.[20] However, these studies concern other regions or rely on population-averaged signals. Whether dorsal CA1 PV+ interneurons change their activity, or which of them carry that activity, during the transition from conditioning to extinction remains untested.

1.4 Aims

This thesis asks whether dorsal CA1 PV+ interneurons contribute to the transition from conditioning to extinction. We addressed three questions: (1) How does PV+ activity change across tEBC conditioning and extinction? (2) When the US is withdrawn, does the identity of PV+ cells carrying activity change relative to non-PV cells? (3) Which LFP frequency bands are associated with learning and extinction within the tone–air-puff window?

To address these questions, we combined single-photon calcium imaging with local field potential (LFP) recording in PV-tdTomato-Cre mice. Conditioning and extinction trials were delivered consecutively in one continuous recording, so that the same PV+ and non-PV cells could be compared on either side of US withdrawal without registering cells across sessions.

2 Materials and methods

2.1 Experimental animals

Sex-balanced PV-tdTomato-Cre transgenic mice (n = 8) were used. Parvalbumin interneurons were identified by endogenous tdTomato fluorescence. All procedures followed institutional animal care and use committee (IACUC) guidelines.

2.2 Surgery and virus injection

All surgeries were performed aseptically under the Institutional Animal Care and Use Committee protocol. Mice (8–12 weeks old) were anesthetized with isoflurane and implanted with a custom chronic imaging-window assembly and a single-channel electrophysiological recording electrode. Anesthesia was induced with 5% isoflurane in 1 L/min O_2 and maintained through a nose cone at 1.5–2.0% isoflurane in 1 L/min O_2. Dexamethasone (2 mg/mL) was administered intraperitoneally at 2 mg/kg before surgery. During surgery, each animal received 0.5 mL saline and buprenorphine (0.5 mg/mL) at 3.25 mg/kg, both subcutaneously.

The imaging window consisted of a stainless-steel cannula (outer diameter 3.17 mm, inner diameter 2.36 mm, height 2 mm) fitted with a circular size-0 glass coverslip (outer diameter 3 mm). A 26-gauge guide cannula (Plastics One) was attached to the window at approximately 45°, with its distal opening aligned to the window surface. A single-channel LFP electrode was integrated alongside the coverslip and guide cannula with UV-curable optical adhesive (Norland Products); its tip extended approximately 1 mm below the coverslip to permit local-field-potential recording.

To access dorsal hippocampal CA1, a craniotomy was centered 2.0 mm posterior and 1.7 mm lateral to bregma (AP −2.0, ML −1.7). The overlying cortex was aspirated to expose the corpus callosum as an anatomical landmark. The white matter was then thinned under a surgical microscope until the hippocampal target was visible. The imaging-window assembly was lowered to the experiment-specific CA1 position and depth, and a ground/reference electrode was implanted in the contralateral hemisphere. Quicksil silicone elastomer was applied around the window perimeter. Metabond dental cement secured the window, guide cannula and custom aluminum head-bar for subsequent head-fixed recordings.

Four weeks after window implantation, pAAV9-Synapsin-jGCaMP8f was infused through the guide cannula into dorsal CA1 to express the neuronal calcium indicator.[21] The infusion volume was 1000 nL at 80 nL/min. This interval allowed the window to settle and the tissue to clear before virus delivery. Imaging began one month after the injection to allow calcium reporter expression. PV identity was determined independently from endogenous tdTomato fluorescence.

2.3 Behavioral paradigm and analysis

2.3.1 Trace eyeblink conditioning paradigm

Mice were head-fixed on a platform inside an enclosure equipped with a speaker for the tone CS, a nozzle for the corneal air-puff US and a camera to record the eye (Figure 1a). The camera recorded the eyelid at 20 Hz. Figure 1a also shows the chronic assembly: the imaging window over dorsal CA1, the guide cannula through which the virus was delivered and the contralateral ground pin. The single-channel LFP electrode is integrated into the window assembly and cannot be resolved separately in this view. Each trial comprised a 350 ms, 9500 Hz tone at 78–84 dB, a 250 ms stimulus-free trace interval and a 100 ms air puff at 5 psi. The US therefore began 600

ms after tone onset. Trials were separated by 19 ± 2 s, yielding a trial period of approximately 20 s (Figure 1c). Timing was verified against the TDT clock. In the session used for alignment QC, the measured tone-to-puff interval was 0.601 s (SD 0.002 s, n = 40 CS+ trials), so the 0–0.6 s response window used throughout ends before US onset.

Animals first received a habituation session with CS-only presentations (Day 0), followed by daily trace-conditioning sessions of 80 paired CS–US trials (Days 1 to N; Figure 1b). On Day N+1, the session analyzed in this thesis comprised 40 paired CS–US trials followed by 40 tone-only trials, in which the air puff was omitted (Figure 1c). This design captured conditioning and extinction in one continuous recording and one imaging field. As in the figures, C1–40 denotes the paired trials and E1–40 the tone-only trials.

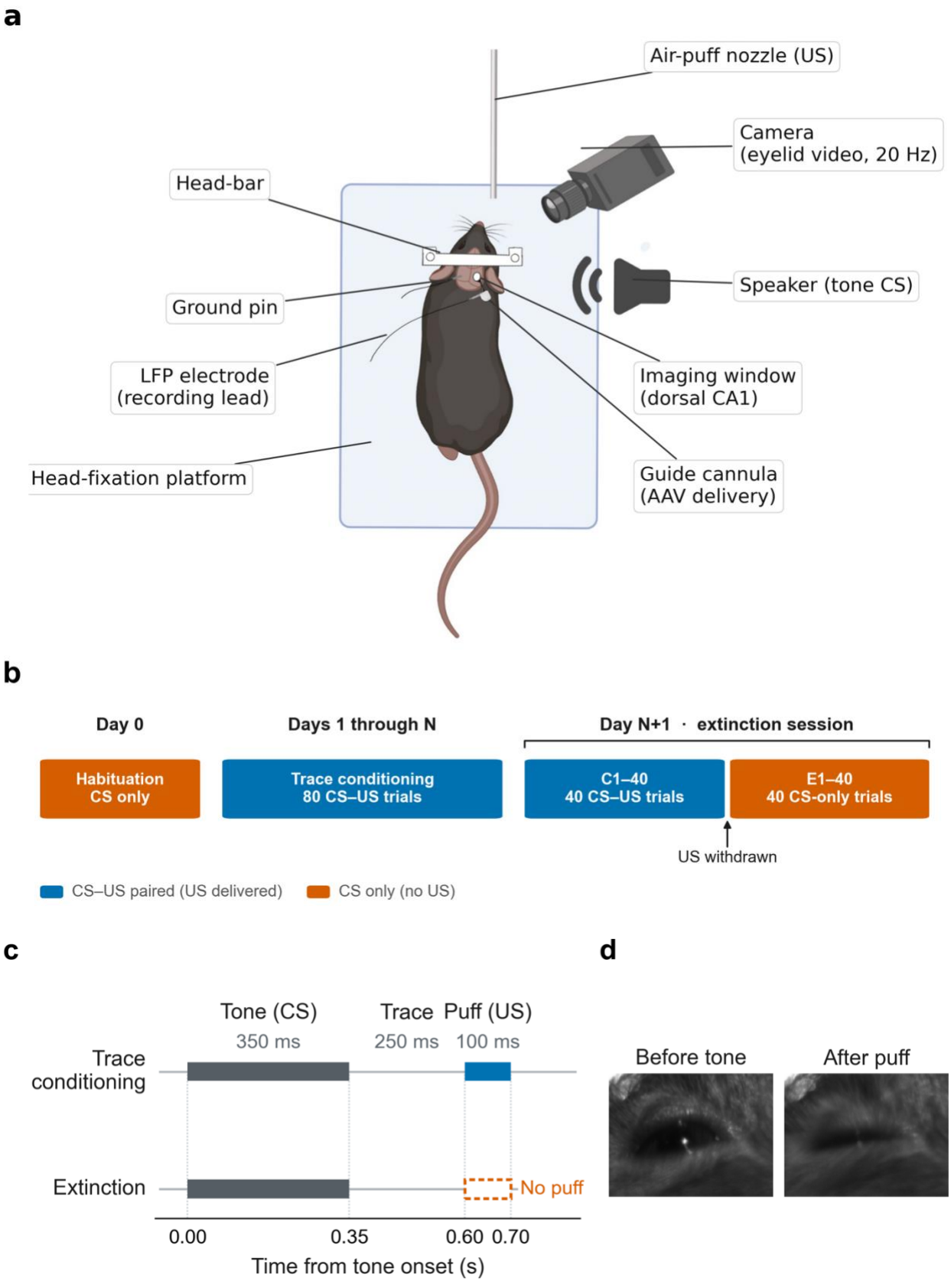

a
Air-puff nozzle (US)
Camera
(eyelid video, 20 Hz)
Head-bar
Speaker (tone CS)
Ground pin
LFP electrode
(recording lead)
Imaging window
(dorsal CA1)
Head-fixation platform
Guide cannula
(AAV delivery)
b
Day 0
Days 1 through N
Day N+1 · extinction session
Habituation
CS only
Trace conditioning
80 CS–US trials
C1–40
40 CS–US trials
E1–40
40 CS-only trials
US withdrawn
CS–US paired (US delivered)
CS only (no US)
c
Tone (CS)
Trace
Puff (US)
350 ms
250 ms
100 ms
Trace
conditioning
Extinction
No puff
0.00
0.35
0.60
0.70
Time from tone onset (s)
d
Before tone
After puff

Figure 1. Head-fixed preparation and trace conditioning paradigm. (**a**) Top view of the recording apparatus and the chronic implant (schematic, not to scale). (**b**) Session timeline. Blue, CS–US paired trials; vermillion, CS-only trials. (**c**) Within-trial structure of the two trial types of the extinction session; dashed box, omitted puff. (**d**) Mean eye-camera images before the tone and after the air puff from a representative trial.

2.3.2 Behavior recording

Eyelid behavior was recorded throughout each session. A synchronization pulse timestamped every video frame on the TDT recording system allowing for alignment of camera triggers to electrophysiological signals. Frame times, tone onsets and air-puff onsets thus share one clock, and video and stimulus timing required no post-hoc alignment. At 20 Hz, each session yielded 36,000 frames. Eyelid closure is visible in the video because the closing eyelid is brighter than the open eye.

2.3.3 Conditioned-response scoring

A conditioned response (CR) was defined as anticipatory eye closure between CS onset and US onset (0–0.6 s). The blink following the air puff was the unconditioned response and was not counted. CRs were scored by hand on the extinction day for every animal (8 animals × 80 trials = 640 trials), with each trial labeled CR or no-CR. These labels were used to calculate the conditioned-response rate. During training, CRs were checked after every trace-conditioning session by manual inspection of the video and by two automated measures of eyelid closure: markerless tracking of the eyelid with DeepLabCut[22] and a grayscale (pixel brightness) measure. Animals were trained for at least five days. Readiness for extinction was then judged from the

CR rate of the last trace-conditioning session (Day N): an animal was moved to the extinction session (Day N+1) only when this rate reached at least 50%.

2.4 Calcium imaging and preprocessing

2.4.1 Image acquisition

Fluorescence was recorded with a multi-channel single-photon microscope. A scientific complementary metal-oxide semiconductor (sCMOS) camera (ORCA-Flash4.0 LT, Hamamatsu) was coupled to a 10× objective lens (N Plan 10×/0.25 PH1, Mitutoyo Plan Apo). The system provided a 1.511 × 1.511 mm² field of view across a 1152 × 1152 pixel frame, corresponding to a spatial resolution of 1.312 × 1.312 μm² per pixel. A commercial single-lens reflex camera lens (Mitutoyo M Plan Apo 10x / 0.28), focused at infinity, served as the tube lens.

Two LED and filter-set combinations were used. For the red channel, a 565 nm LED (M565L3, 880 mW minimum output, Thorlabs) was paired with a DsRed filter set (49005, Chroma; excitation 545/30 nm, dichroic T570lp, emission 620/60 nm). For the green channel, a 470 nm LED (M470L4-C1, 420 mW typical output, Thorlabs) was paired with an EGFP filter set (49002, Chroma; excitation 470/40 nm, dichroic T495lpxr, emission 525/50 nm). Images were acquired using HCImage software (Hamamatsu Photonics) at 20 Hz.

Short red-channel videos were recorded to capture endogenous tdTomato fluorescence. Cell type was assigned by cross-registering ROIs in this channel with the GCaMP ROIs.

2.4.2 Motion correction

Each movie was motion-corrected by rigid registration in CaImAn,[23] which implements the rigid stage of NoRMCorre.[24] For each frame, a single global translation was estimated at subpixel

precision from the peak cross-correlation with a template computed from that movie. Pixels exposed at frame edges were filled by replicating the border because the DeepWonder networks cannot process undefined values.

2.4.3 DeepWonder

Wide-field single-photon recordings contain substantial out-of-focus background from scattered fluorescence. Because paired wide-field and background-free recordings were unavailable as training ground truth, DeepWonder[25] was used. This method is trained on realistic simulated recordings with known background-free frames and then applied to experimental data. It was used to remove background and extract each neuron's spatial footprint and activity trace. The released pretrained RB-Net and NS-Net were used; they had been trained on NAOMi1p simulations of the RUSH system.[25] The motion-corrected movie was divided into overlapping blocks of 304 × 304 pixels × 2000 frames, with 32-pixel spatial and 40-frame temporal overlap. Each block was processed separately, and outputs were stitched together after the overlap margins were discarded. Processing proceeded in four stages.

The Removing Background Network (RB-Net) first generates a background-free movie, so that out-of-focus fluorescence does not mix into the signal of individual cells. This stage removes background contamination but not photobleaching; session-long bleaching is corrected separately (§2.4.4). To locate neurons, the Neuron Segmentation Network (NS-Net) then processes the background-free movie in 64-frame windows and returns one two-dimensional neuron-probability mask for each window. The network architectures are described in Zhang et al.[25].

After that, the window masks were converted into neuron candidates and classified, so that only soma-shaped regions were kept as neurons. Each mask was binarized at 30% of its maximum value and divided into 8-connected components. A segment that overlapped another spatially but appeared in a different window was treated as a separate candidate. Components whose outline contained a hole (more than one contour) were discarded. For each remaining component, roundness was calculated as

$$\Theta = \frac{4\pi s}{p^2},$$

where s is the area and p the perimeter of the component.

Candidates with $\Theta \leq 0.5$ or an area of 60 pixels or less (≈103 μm² at the 1.312 μm pixel size of these recordings) were discarded. Candidates with $\Theta \geq 0.8$ were assigned to the 'good' list and treated as single neurons; those with $0.5 < \Theta < 0.8$ were assigned to the 'bad' list.

Candidates were then grouped into neurons. Two candidates were assigned to the same neuron if their centroids were within 20 pixels and either their traces correlated at $r > 0.9$ or their footprints overlapped by more than 90%. For this comparison, each trace was the mean of 10 randomly sampled footprint pixels. Within each group of good candidates, only the largest footprint was retained. A good neuron smaller than 100 pixels was removed if it overlapped a nearby neuron by more than 20%. Within each group of bad candidates, footprints were combined into their union. Pixels belonging to any good neuron were then removed from this union, and the largest remaining connected region was kept as an additional neuron if its area exceeded 200 pixels and its roundness exceeded 0.5. Each group of bad candidates therefore contributed at most one neuron. The additional neurons were grouped using the same criteria, and their footprints were

combined into a union. Where two good or additional neurons overlapped by more than 60%, only the larger neuron was retained, so that the same cell was not counted twice.

Finally, each neuron's activity was calculated as the mean signal across all pixels in its footprint in the background-free movie. Shared pixels were included, so neurons with overlapping footprints may contain some signal from one another.

2.4.4 Photobleaching correction (airPLS)

During continuous imaging, the excitation light gradually bleaches the fluorescent indicator, so the same neural activity produces a smaller signal late in a session than early in it. Because conditioning and extinction trials were recorded in sequence, this slow decline could be mistaken for a change between the two phases. In these recordings, evoked responses declined by roughly 12% over a session, and the decline did not follow a simple exponential time course. Each cell's slow baseline was therefore estimated with a flexible, non-parametric method, adaptive iteratively reweighted penalized least squares (airPLS),[26] as implemented in pybaselines.[27] The baseline was estimated after decimation (20 Hz → 2 Hz) with smoothing parameter $\lambda = 10^7$ and then interpolated to the original frame rate.

Because bleaching lowers both the baseline and the evoked responses, each trace was divided by its normalized baseline, which restores the lost gain. The divisor was floored at 20% of the baseline median, limiting the restored gain to 5× in cells with baselines near zero. The correction was applied before all downstream analyses, so population labels and later measures were based on the same corrected signal.

2.5 Population and ensemble analyses

Cells were first divided into PV+ and non-PV populations. PV+ cells were identified by manually annotating the tdTomato channel and cross-registering the labels to the GCaMP ROIs; inspected cells without a PV+ label formed the non-PV population. For the CO/EX classification, only cells with a degenerate baseline median absolute deviation (MAD $\leq 10^{-6}$) were excluded. For the recurrence analysis, cells also had to pass a pre-specified signal-quality gate on the raw trace (noise level, first-difference autocorrelation and spectral criteria). The gate was applied identically to PV and non-PV cells and used only C1-20 trials, which precede every block compared, so it could not select cells on the effect being measured. Non-PV cells with a tdTomato z-score above 2.9 were excluded as possible unlabeled PV+ cells.

The PV comparisons used the four animals that combined clear conditioned responding (CRs on 23–30 of the 40 trace-conditioning trials) with the largest cell samples (a minimum of 11 eligible PV cells and more than 370 non-PV cells in every session). PV cells were compared with 100 random samples of the same number of non-PV cells (equal-N draws), each kept fixed across blocks. Group statistics used the animal as the unit: two-sided one-sample t tests and exact sign-flip tests on the per-animal values. Because baseline MAD varied about tenfold across animals, MAD-scaled amplitudes were not compared across animals.

The 80 trials of the extinction session were divided into eight 10-trial blocks (conditioned: C1–10 to C31–40 and extinction: E1–10 to E31–40). For analysis centered on US withdrawal, we compared C32-40 with E2-10 which consisted of only 9 trials. This allowed us to exclude trial E1, which represents the first unexpected extinction trial as well as the removal of corresponding single trial C31 from the conditioning session to balance both blocks at 9 trials each.

A non-PV cell was classified as conditioning-responsive (CO) if its mean fluorescence over 0–0.6 s after tone onset, averaged across the 40 conditioning trials, exceeded a circular-shift null

(1,000 random shifts of the trace relative to trial onsets; one-sided $p < 0.05$, uncorrected across cells). The same test on the 40 extinction trials identified extinction-responsive (EX) cells. Shifting the whole trace keeps each cell's autocorrelation and slow drift while removing alignment to the tone. The partition was adapted from Mount et al.[10] with four changes: responses were mean fluorescence in place of detected calcium events; the 0–0.6 s window ended before the US, so a label could not reflect the puff or its absence; all 40 trials of each phase were used, compared with 20; and the null shifted the whole trace.

Identity recurrence asks whether the same cells carry high relative activity from trial to trial within a block, which a block-averaged pattern cannot show. On each trial, a cell's response was its mean fluorescence over 0–0.6 s was normalized to the pre-tone period (mean over −1 to 0 s). To account for changes that could be driven by photobleaching across the trial, a stimulus-free control window (−3 to −2.4 s minus −4 to −3 s), was also included as an additional normalization step. Responses were centered and scaled by each cell's median and 1.4826 × MAD over C1–20 (scale floored at 10% of the median positive MAD across eligible cells) and clipped at ±3. The component along the uniform-fluorescence axis (a unit vector proportional to the reciprocal of each cell's scale) was then projected out, so that a gain change shared by the whole population could not alter cell ranks.

On each trial, the top 20% of cells by response (rounded up; K of the N cells) formed the high-activity set; because its size was fixed, the measure reflects which cells carry the activity, not how much. The trials of each block were split into interleaved odd and even halves, and the Jaccard index $J = |A \cap B|/(2K - |A \cap B|)$ was averaged over all cross-half trial pairs (20 pairs for nine-trial blocks, 25 for ten-trial blocks) and corrected for chance as

$$R = \frac{\bar{J} - \mathbb{E}[J]}{1 - \mathbb{E}[J]},$$

where $\mathbb{E}[J]$ is the exact hypergeometric expectation of J for two random K-subsets of N cells, so that $R = 0$ is chance-level recurrence. Recurrence was computed for the PV cells and for each equal-N draw, and PV enrichment was defined for animal a and block w as

$$E_R(a, w) = [R_a^{\mathrm{PV}}(w) - R_a^{\mathrm{PV}}(\mathrm{C32–40})] - \mathrm{median}_m[R_a^m(w) - R_a^m(\mathrm{C32–40})],$$

where m indexes the 100 equal-N draws. $E_R < 0$ means that PV recurrence declined more than that of matched non-PV cells.

2.6 Local field potential (LFP)

2.6.1 LFP recording

Local field potentials were recorded using an insulated stainless-steel electrode (P1 Technologies, 7N003736501F) integrated into the imaging-window assembly, with a ground/reference electrode in the contralateral hemisphere. Signals were acquired using a Tucker-Davis Technologies RZ10x processor and stored at 6103.5 Hz, alongside stimulus and synchronization channels used to timestamp imaging and video. Because recording was single-channel, the analysis cannot provide a laminar profile, current-source density or inter-channel coherence. All LFP analyses used the four animals of the PV comparisons and three sessions from each: the extinction session (trace-conditioning trials C1–40 and extinction trials E1–40), the last trace-conditioning session before it (80 CS–US trials with the same schedule) and the naive habituation session (20 tone trials without US, 26 s apart). Trial onsets were extracted from the recorded stimulus events using the same onset convention as in the calcium analyses.

2.6.2 LFP preprocessing

Each trial epoch (−10.5 to +2.5 s relative to tone onset) was preprocessed in a fixed order. A fourth-order Butterworth low-pass filter at 150 Hz and an IIR notch filter at 60 Hz (Q = 30) were first applied, both with zero phase. The signal was then downsampled to 500 Hz using polyphase filtering. Results are reported for the signal after this preprocessing. On trace-conditioning trials, the signal after the air puff was dominated by the puff artifact and was not analyzed. For time–frequency maps, each trial's signal from 0.58 s onwards was replaced by its mirror image around 0.58 s, preventing post-US data from entering any wavelet. Window means differed by less than 0.05 dB from those of the unmirrored maps.

2.6.3 Spectral estimation and frequency bands

Time–frequency maps were calculated using Morlet wavelets (2–100 Hz, 40 logarithmically spaced frequencies, 3–10 cycles[28]). Frequency bands followed published definitions and were fixed before the present analyses; none was selected from the data (Table 1).

Table 1. LFP frequency bands. Band limits follow published definitions of rodent brain rhythms: delta,[29] theta, slow (low) gamma and fast gamma,[30] and beta.[31] Three limits were adjusted: delta starts at 2 Hz, the lowest wavelet frequency; low gamma starts at 30 Hz, where beta ends; and fast gamma starts at 62 Hz to stay clear of the 60 Hz notch. Alpha (12–15 Hz) covers the interval between theta and beta. The tone-onset region (9–12 Hz, 0.03–0.17 s) was identified from the data and is not a predefined band.

Band	Limits
Delta	2–4 Hz

Band	Limits
Theta	4–12 Hz
Alpha	12–15 Hz
Beta	15–30 Hz
Low gamma	30–55 Hz
Fast gamma	62–100 Hz

2.6.4 Tone-onset power across sessions

This analysis used the four animals of the PV comparisons and signals after preprocessing. For each trial, Morlet power was expressed as the change from the trial's own mean pre-tone log power (−1.0 to −0.1 s), and every trial was mirrored at 0.58 s. For each animal, the median map of the 40 C trials was subtracted from that of the 40 E trials, and the differences were tested across animals with a cluster-mass test (exact sign flips[32]). The test was restricted to points that no wavelet of the C trials could reach from the air puff, defined as 0.601 s minus three temporal standard deviations of the wavelet at each frequency. In the signal without comb removal, the strongest region before this limit lay at 9–12 Hz and 0.03–0.17 s after tone onset. This region was then fixed and used for all further tests, including those on other sessions. The region value of a trial was its mean power change within the region.

Trials were grouped into blocks of 20 (C1–20, C21–40, E1–20 and E21–40 on the extinction session; T1–20 to T61–80 on the preceding trace-conditioning session; N1–10 and N11–20 on the naïve habituation session). Each block was summarized by its median with a bootstrap 95% CI (2,000 resamples). Time within the session was controlled using the preceding trace-conditioning session, which had the same 80-trial schedule and tone but no US withdrawal. For

each session, the median of trials 41–80 minus that of trials 1–40 was calculated; the extinction-session value minus the preceding-session value isolated changes not attributable to time in the session. The same contrast was calculated at every time–frequency point, and for the pre-tone level of each band of Table 1 (−0.5 to −0.1 s), expressed relative to one pre-tone reference per session. To locate the change in frequency, the same contrast was also measured in each band of Table 1 in its own time window. This window was the contiguous period, within the band's puff-free time after tone onset, around the band's largest mean change, over which the mean stayed above half of that peak. To keep the choice of the window separate from the test, the window was chosen from three animals, and the fourth animal was measured in it, for each animal in turn. Results are reported as the mean across animals and the number of animals with the same sign.

3 Results

3.1 Conditioned responding is reduced when the US is withdrawn

Before asking how CA1 activity changed at the boundary, we first determined whether withdrawing the US produced behavioral extinction within the session. Conditioned responses were scored by hand on every trial. Hand-scored CR rates were 48.8% in C21–30 and 57.5% in C31–40, then fell to 37.5% in E1–10 and 33.8% in E31–40 (means across eight animals). The decrease from C31–40 to E1–10 occurred in all eight animals (exact sign-flip $p = 0.0078$), with little further change during extinction. Behavioral extinction was therefore rapid but partial on this day. CRs also alternated frequently between neighboring trials: each animal had CRs on 8–30 of the 40 trace-conditioning trials and 1–27 of the 40 extinction trials.

3.2 PV+ interneurons and non-PV neurons in calcium imaging

Comparing PV+ and non-PV cells across the boundary required both populations to be resolved in the same field of view. Calcium activity was imaged through the chronic dorsal CA1 window in all eight animals. Figure 2 shows the extinction-session field of view from an example animal. jGCaMP8f fluorescence revealed densely packed somas, while endogenous tdTomato marked a sparse set of PV+ somas (Figure 2a). ROIs were extracted with DeepWonder, and PV identity was assigned by manual annotation of the tdTomato channel. In this animal, 31 ROIs were labeled PV+ (magenta in Figure 2a). In the enlarged 262 × 262 µm region, all six PV masks overlap tdTomato-positive somas (Figure 2b). The masks shown are DeepWonder footprints expanded to each soma's half-maximum contour (at most 5 pixels). They were defined on the GCaMP image, and a residual offset of about 2 pixels relative to tdTomato was not corrected. PV+ and non-PV cells could therefore be followed in the same continuous recording.

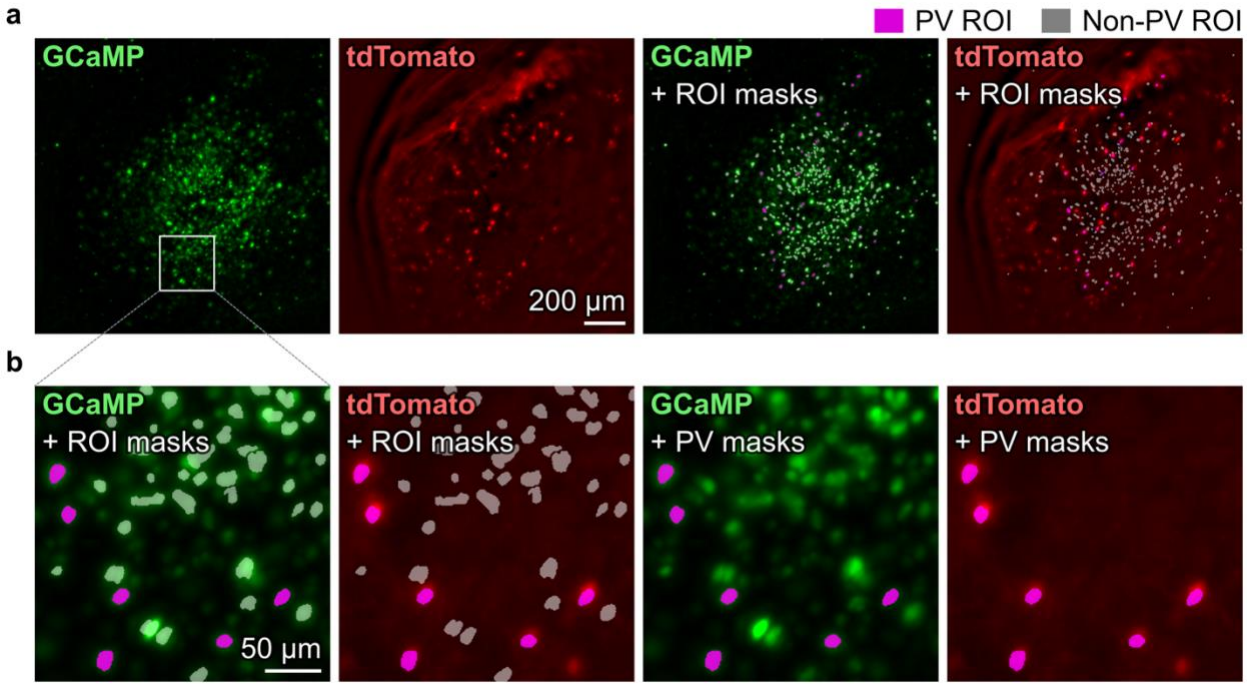


Figure 2. Field of view and ROI identity in an example animal (extinction session). (**a**) Whole field of view (1152 × 1152 pixels, 1.511 × 1.511 mm). GCaMP, maximum projection of the

background-removed movie; tdTomato, mean of 300 motion-corrected frames. Magenta, PV ROIs; gray, non-PV ROIs. Box, region enlarged in b. (**b**) Enlarged region with all ROI masks or PV masks only. Both channels use the same linear display limits (0.5th to 99.9th percentile) in a and b. Scale bars, 200 μm (a) and 50 μm (b). All labeled ROIs are shown, before the inclusion criteria applied in later analyses (§2.5).

3.3 CO and EX cells reverse their activity after US withdrawal

Mount et al. reported that conditioning and extinction recruit distinct CA1 populations.[10] To describe how such populations behave across the boundary in the present recordings, we applied an adapted version of that partition. Figure 3 illustrates it in an example animal, which showed the clearest separation among the four animals used for the PV comparisons. In this animal, 51 non-PV cells were classified as CO and 128 as EX. Seven cells belonged to both sets, compared with 5 expected if CO and EX membership were independent given the observed set sizes.

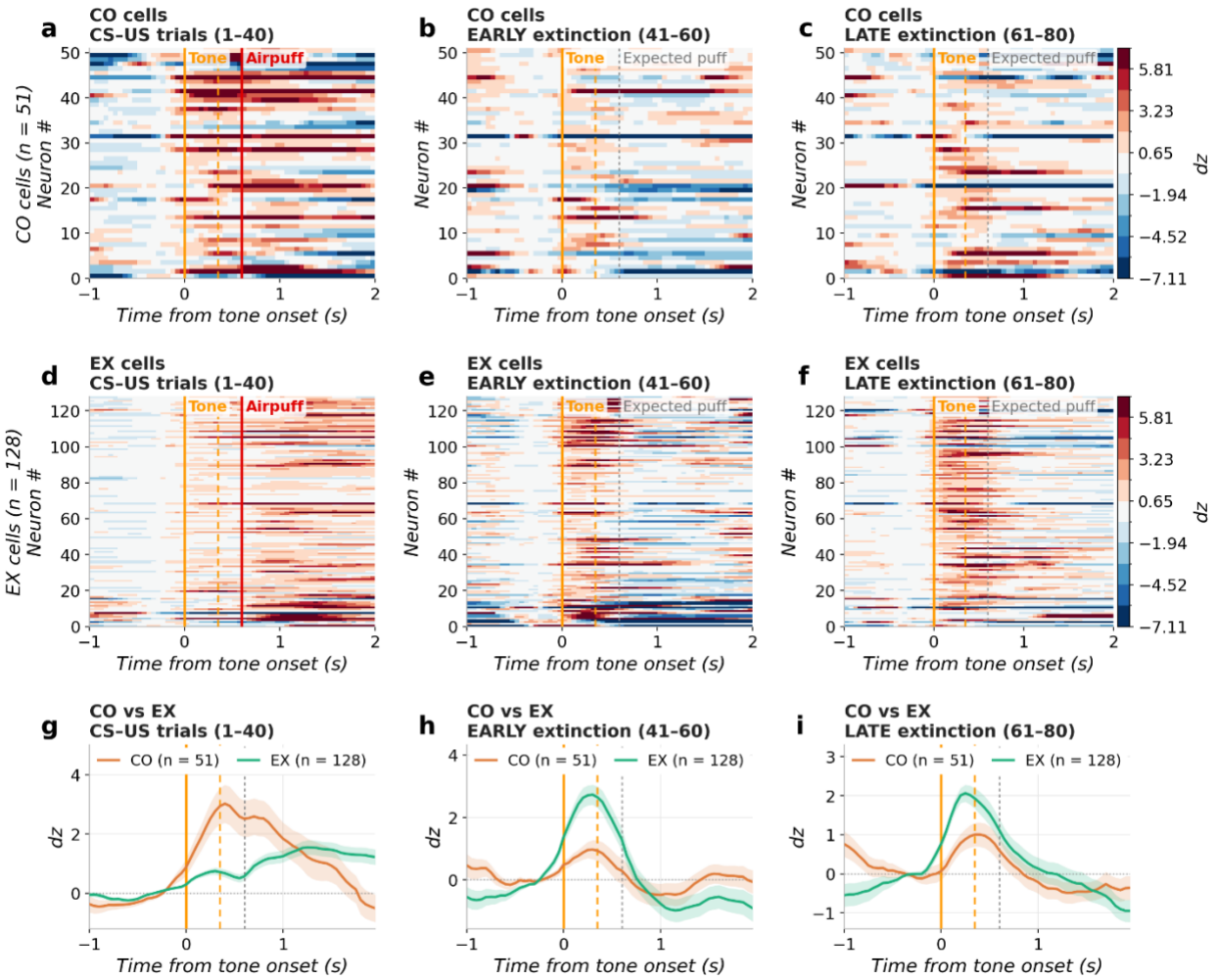


Figure 3. CO and EX response profiles across the extinction session from an example animal. Columns are conditioning (C1–40), early extinction (E1–20) and late extinction (E21–40). (**a**–**c**) CO cells and (**d**–**f**) EX cells; each row is one cell, in the same order in all three columns. Heatmaps show even-numbered trials only. (**g**–**i**) CO and EX population means ± SEM over all trials of each period. Orange solid and dashed lines, tone onset and offset; red line, air puff; gray dotted line, expected puff. Color, MAD z-score (dz).

CO cells responded strongly during conditioning and less during extinction (CS→US window means: 2.27 dz during conditioning, 0.73 during early extinction and 0.72 during late extinction; Figure 3a–c). In the heatmaps, the tone-locked activity of CO cells seen during conditioning was

much weaker after US withdrawal. EX cells showed the opposite pattern. They responded weakly during conditioning and more strongly after US withdrawal (0.59 dz during conditioning, 2.22 dz during early extinction and 1.62 dz during late extinction; Figure 3d–f), and their response was largest during the first 20 extinction trials. The population means show the reversal directly: during conditioning the CO mean exceeded the EX mean, whereas from the first extinction block onward the EX mean exceeded the CO mean (Figure 3g–i).

The heatmaps show only the even-numbered trials, and row order was set on the odd-numbered trials. Early and late extinction are therefore averaged over the same number of trials, so neither is smoother for reasons unrelated to activity. Population membership, however, was determined from all 40 trials of each phase. The within-phase profiles (CO during conditioning and EX during extinction) are therefore in-sample descriptions, and a population selected for a large response in one phase may regress toward the mean in the other. Because all conditioning trials preceded all extinction trials, a change between phases could also reflect time in the session.

3.4 Active PV+ cells recur less consistently at early extinction

Block averages cannot distinguish a stable active population from one whose members alternate across trials. Accordingly, trial-to-trial recurrence of high relative activity among PV cells was assessed within each block. Identity recurrence R was calculated for each population's high-activity set (chance level: $R = 0$), and PV enrichment E_R was measured relative to 100 fixed equal-N non-PV draws (§2.5). The analysis included 11-28 PV cells and 370-1223 non-PV cells per animal that met the inclusion criteria (§2.5).

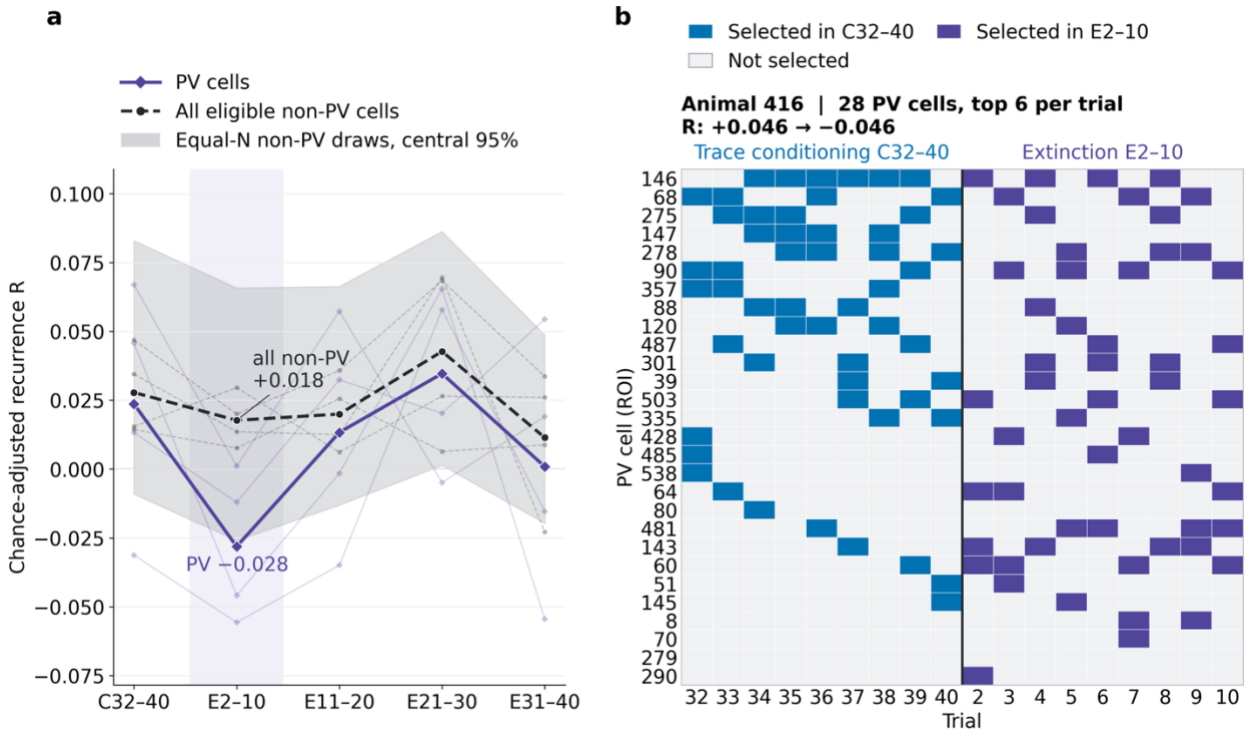


Figure 4. PV identity recurrence across US withdrawal (n = 4 animals). (**a**) Identity recurrence *R* of the high-activity set for PV cells (purple) and all eligible non-PV cells (dark gray), from C32–40 through E31–40. Thin lines, animals; thick lines, means of animals; gray band, range expected from equal-N non-PV draws. $R = 0$ is chance. (**b**) Trial-wise high-activity membership of the PV cells of an example animal across the boundary. Rows, cells; columns, trials (blue, C32–40; purple, E2–10); filled tiles, the top 20% of most active cells on each trial (corresponds to N = 6 per column for the example shown).

Before the boundary, recurrence was similar for PV and non-PV cells (C32–40: PV +0.024; all non-PV cells +0.028). The PV-minus-equal-N non-PV difference was −0.004 (2/4 animals negative; t-test p = 0.86). In E2–10, PV recurrence fell to −0.028, while the non-PV pool remained at +0.018 (Figure 4a). In Figure 4a, the gray band marks the central 95% interval of the across-animal mean over the 100 equal-N non-PV draws. The changes from baseline were

−0.052 for PV cells, −0.010 for all non-PV cells and −0.012 for equal-N draws. The resulting PV enrichment, E_R, was −0.040 and negative in all four animals (t-test p = 0.063; exact sign-flip p = 0.125). Absolute PV recurrence in E2–10 was also lower than recurrence in equal-N non-PV draws (−0.046, 4/4; t-test p = 0.036). In later blocks (E11–20 to E31–40), E_R ranged from −0.005 to −0.001, with 2/4 animals negative in each block. The PV-enriched reduction in recurrence was limited to the first extinction block.

Figure 4b displays the cell memberships underlying this statistic in an example animal with 28 eligible PV cells. Each column marks the six cells with the highest relative activity on that trial, and rows are ordered by how often each cell was selected in C32–40, so the extinction trials play no part in the order. In C32–40, the same cells were selected on several trials; in E2–10, selections were spread across more cells, and R fell from +0.046 to −0.046. The results show that PV cells that repeatedly carried high relative activity in C32–40 were replaced by a less consistent set in E2–10, and this recurrence changed from C32–40 to E2–10 occurs in all four animals.

At extinction onset, PV cells showed a larger, transient reduction in trial-to-trial identity recurrence than matched non-PV cells. This result does not imply that other aspects of PV activity were static.

3.5 The tone-onset LFP response grows over extinction

PV identity recurrence changed in the first extinction block. To ask whether the CA1 LFP also changed after the US was withdrawn, the LFP recorded in the same sessions was examined in the four animals used for the PV comparisons.

Before the air puff could enter any wavelet, the strongest difference between E and C trials in these animals lay at tone onset, at 9–12 Hz and 0.03–0.17 s (E > C in 4/4 animals; cluster $p = 0.125$, the smallest value attainable with four animals). This region was then fixed. After comb removal, E − C in the region was +1.1 to +3.6 dB in all four animals.

The increase did not appear as an abrupt step at the first omitted US; it built up over the extinction trials. Across blocks of 20 trials, the region value rose from −1.3 dB in C1–20 and −0.5 dB in C21–40 to +0.9 dB in E1–20 and +2.6 dB in E21–40 (means of four animals; Figure 5c). In E21–40 it was positive in all four animals (+1.2 to +3.3 dB; bootstrap 95% CIs excluding zero, lowest lower bound +0.3 dB). The change at the boundary differed between animals (E1–20 minus C21–40: +0.6 dB in 415, −2.5 dB in 416, +5.0 dB in 417 and +2.3 dB in 420). The largest step, in animal 417 (+4.1 dB beyond the linear trend, 95% CI 0.9 to 7.4), was not unique: steps of similar size occurred at trial 41 of the preceding conditioning session, where nothing changed (up to +2.7 dB, 95% CI 0.4 to 5.0). Single-animal steps therefore lie within the variability of the measure.

Because all C trials preceded all E trials, the rise could have reflected time within the session. The preceding trace-conditioning session, with the same 80-trial schedule and tone but no US withdrawal, controlled for this. On that session the later trials did not exceed the earlier ones (trials 41–80 minus 1–40: −0.1 dB, 2/4 animals positive; trials 61–80 minus 1–20: −1.7 dB, 0/4; linear trend −0.16 dB per 10 trials, 0/4). On the extinction session the same contrasts were +2.5 dB (4/4), +3.9 dB (4/4) and +0.40 dB per 10 trials (4/4) (Figure 5d). The difference between the two sessions was +2.6 dB (4/4 animals), and +3.0 dB (4/4) in the signal without comb removal.

Applied at every time–frequency point, the same contrast (per animal, E − C on the extinction session minus trials 41–80 minus trials 1–40 on the preceding session, averaged across the four

animals) placed the change at tone onset, inside the 9–12 Hz region (Figure 5a). Measured in its own time window, chosen from the other three animals, the 9–12 Hz response increased in all four animals; theta (4–12 Hz, which contains 9–11 Hz) and beta increased in three of four, and alpha, low gamma and fast gamma did not change consistently. Delta could not be measured after tone onset, because its wavelets reach the air puff before the tone. No band changed consistently before the tone. The pre-tone alpha level was higher on E than on C trials (+0.8 dB, 4/4), but the second half of the preceding session showed a similar increase (+0.7 dB, 3/4), indicating drift within the session.

Figure 5b shows the time course of 9–12 Hz power in each session. On the naive day, the tone-onset response at 9–12 Hz was −0.2 dB in the first ten trials and −0.6 dB in the last ten. Late extinction exceeded this level (E21–40 minus naive: +2.6 dB, 3/4 animals), whereas the C trials of the extinction session did not (−0.7 dB, 2/4; Figure 5b, c). Naive and extinction trials were both tone-only trials without an air puff; they differed in that the extinction trials followed conditioning. Comparisons between sessions are less reliable than contrasts within a session: C1–40 differed from the last 40 trials of the preceding session by −1.2 dB (0/4 animals positive; up to −3.2 dB in one animal). Repeated tone-only trials alone did not produce the increase. Across the first 20 tone-only trials of a session, the response did not rise consistently on the naive day (trials 11–20 minus 1–10: −0.3 dB, 2/4) or in extinction (E11–20 minus E1–10: −0.8 dB, 2/4).

Thus, in the four PV animals, the tone-onset LFP response at 9–12 Hz increased after US withdrawal, grew over the extinction trials and in late extinction exceeded its naive level. This change was not explained by time within the session or by repetition of the tone. Because tone-only trials on the naive day, which also lacked the air puff, did not show it, the naive comparison

argues against the absence of the puff as such, although comparisons between days are less reliable than those within a session. The change is therefore associated with the extinction procedure, in which an expected air puff is omitted. Whether it reflects extinction learning cannot be determined here, because conditioned responding changed little after the first ten extinction trials. Each result rests on four animals (smallest attainable exact p = 0.125) and on a region identified in their data.

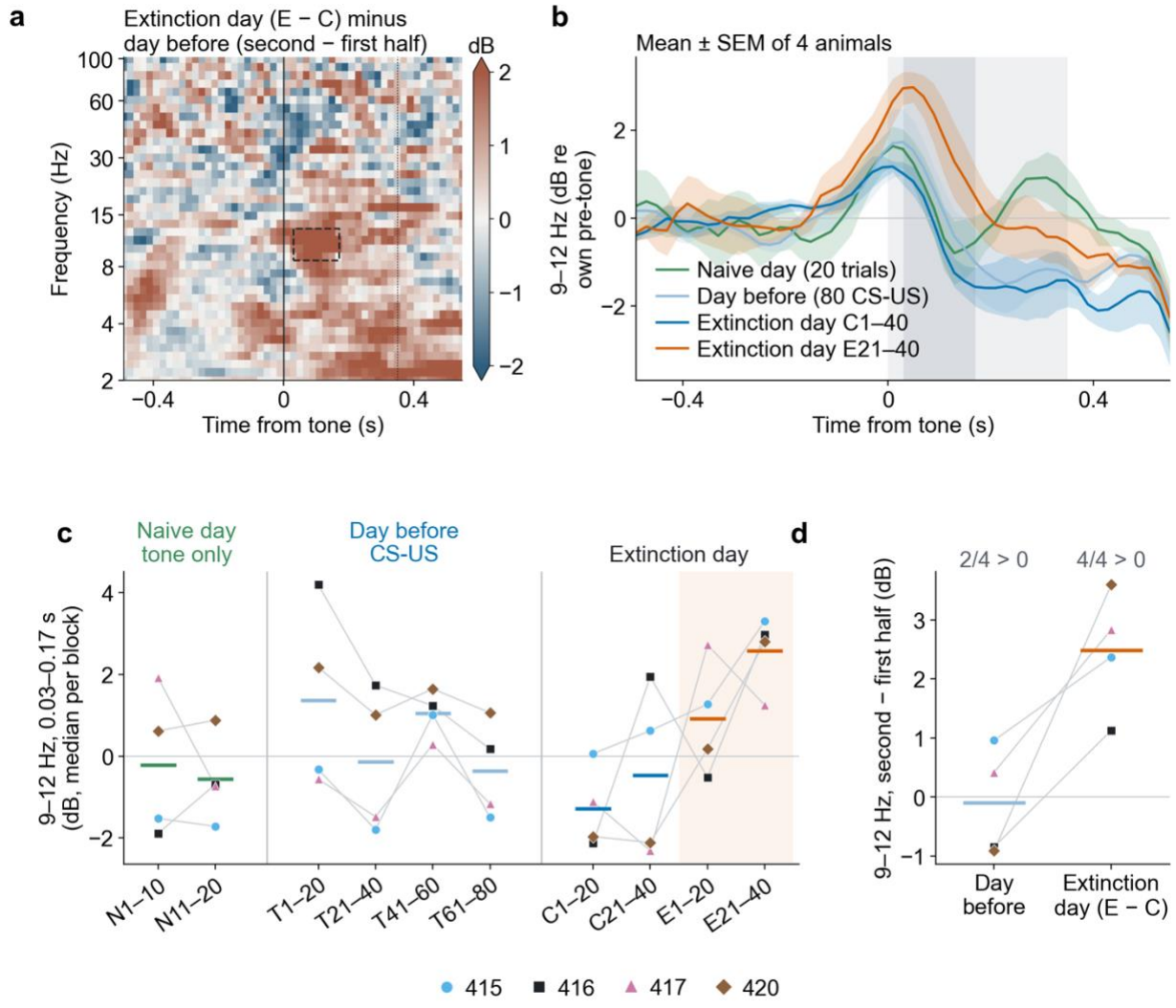


Figure 5. Tone-onset LFP power across sessions in four animals. Power is the change from each trial's own pre-tone power. (**a**) Change over the extinction session beyond that over the preceding session, at each time and frequency; mean of four animals. Dashed box is the tone-onset region (9–12 Hz, 0.03–0.17 s); solid line: tone onset; dotted line: tone offset. (**b**) Power at 9–12 Hz on the naive day, on the preceding session and in C1–40 and E21–40 of the extinction session; mean ± SEM. Dark gray: region window; light gray: tone. (**c**) Region value per block (median over the block's trials); symbols: animal ID; bars: means. (**d**) Trials 41–80 minus trials 1–40 on the preceding session and on the extinction session (E − C); lines join the same animal; above, animals with a positive value.

4 Discussion

In dorsal CA1, withdrawing the US in an established trace eyeblink task was followed by two changes with different time courses. At extinction onset, the PV+ cells with the highest mean activity became less consistent from trial to trial compared to activity matched non-PV cells in all four animals tested, although the difference did not reach significance and depended on how the response was defined. This change was confined to the first extinction block, matching the first omission of the US, but returned to pre-extinction levels after the first 10 extinction trials. The tone-onset LFP response at 9–12 Hz, in contrast, grew over the extinction trials, beyond the change over the same trials of the preceding conditioning day. These results suggest that the omission of an expected outcome briefly redistributes activity among PV+ cells, while a slower change in the tone-evoked CA1 network builds up as extinction learning continues.

Extinction does not erase a learned association; it adds a competing memory, and the hippocampus helps decide which of the two is expressed.[1–4] CA1 is thought to carry this update at the population level,[10,11] and PV+ interneurons, which control which pyramidal cells fire and when,[12,15] are well placed to shape it. Earlier links between PV+ interneurons and extinction came from ventral CA1 and medial prefrontal cortex,[19,20] which differ from dorsal CA1 in connectivity and function.[33] In dorsal CA1, PV+ activity supports early trace eyeblink acquisition and wanes once responding reaches asymptote.[14] The present data suggest that, after learning, PV+ involvement may depend more on the change in the network of neurons encoding the CS, then the overall activity of the PV population. Hippocampal oscillations have likewise been related to learning rate, conditioned responding and learned cues.[31,34–38] The 9–12 Hz change observed here began at tone onset and grew while conditioned responding changed little, which suggests that it reflects a change in what the tone signals once it no longer predicts the air puff. Because PV+ interneurons contribute to pacing theta,[18] the early PV+ change and the later 9-12 Hz could involve a shared mechanism, although the present data do not test this.

These observations depended on following the same identified cells across the boundary within one continuous recording. Earlier imaging of CO and EX populations required co-registering cells across separate recordings in some animals,[10] and PV-targeted fiber photometry averages over the whole population[19]; neither approach can detect a change in which PV+ cells are active. Separating the identity of active cells from their overall activity showed that PV+ involvement at extinction onset, if confirmed, would be invisible to measures of mean activity. The results therefore identify the first trials after US withdrawal as the window in which to test whether hippocampal inhibition shapes how readily extinction is learned, a question relevant to exposure-based therapies that rely on extinction.[2]

Limitations and future directions

The PV result rests on four animals, selected for clear conditioned responding and the largest PV samples, and it depended on the response definition. With four animals, an exact sign-flip test did not fall below $p = 0.125$. The next step is therefore a confirmatory study in more animals with sufficient PV+ cells, applying the recurrence analysis exactly as specified here. The cohort was also sex-balanced, but differences between the sexes were not examined; a larger cohort would allow analyses stratified by sex.

The study is correlational and cannot show that PV+ cells are needed for extinction. Optogenetic suppression or activation of dorsal CA1 PV+ cells around US withdrawal would test whether PV+ activity contributes to the change in recurrence and to the decline in conditioned responding.

Wide-field single-photon imaging limits cell-type and temporal resolution. The PV-tdTomato label covers basket, axo-axonic and bistratified cells, which target different parts of the pyramidal cell and could change in opposite directions.[12] PV+ somas are sparse within the dense pyramidal layer, footprints can share pixels and overlapping candidates were not demixed, so PV traces may contain signal from neighboring cells. Calcium imaging at 20 Hz also cannot resolve the spike timing and gamma-band synchrony through which PV+ basket cells act.[12,15] Cellular-resolution, subtype-specific or PV-targeted recordings at higher temporal resolution would address these limits and confirm that the effect originates in PV+ cells.

Behavioral extinction was rapid but partial: conditioned responding fell when the US was withdrawn and changed little afterwards. The present results therefore describe what happens when an expected air puff stops, not a confirmed change in what the animal has learned.

Following the same cells over several extinction days, with a retention test, would separate the response to omission from extinction learning itself.

The LFP was recorded with a single electrode, so it gives no laminar or spatial information. The tone-onset analysis was not specified in advance, used a region identified from the same animals' data and compared recordings from different days. Multi-site recordings and a pre-specified test in new animals are needed. Because the LFP and calcium signals were recorded simultaneously, relating each cell's trial-by-trial calcium response to the tone-onset LFP would also show whether specific neurons, including PV+ cells, carry activity that is detectable in the LFP. Finally, representing each trial as a point in neural state space could show whether the extinction state moves along the conditioning axis or along a new one, and whether PV+ activity aligns with the dimensions that change.

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Bo Yuan designed the analysis pipeline for PV recurrence presented in Figure 4. He also led the surgery procedures, which I assisted with before independently performing surgeries on a subset of the animals. Besides, he also assisted with behavioral data collection. Throughout the project, the choice of analytical approaches, statistical methods, and interpretation of results were developed in discussion with Dr. Gritton and Bo Yuan.

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